

The Shore Pod Standard

Rev. Alpha

9/29/2025

Amber Skotak, Bradley Denby

Revision History

Table 1: Revision History

Version	Date	Author	Changes
Alpha	2024-12-29	Amber Skotak, Bradley Denby	Initial draft

Contents

Contents

Revision History	2
Contents	3
Introduction	4
Purpose	4
Shore Pod Specification	4
Mechanical Requirements	4
Electrical Specifications	8
Operational Specifications	8

Introduction

The purpose of the Shore Pod was to reduce cost and development time and increase access to ocean sensing, and, in turn, the blue economy. A Shore Pod is a sensor platform buoy that adopts a standard size and form factor, which unit is defined as ‘P’. A 1P Shore Pod is an 8cm (OD) x 12cm (h) cylinder with a total mass of up to 500g. The primary objective of this standard is to provide specifications for the design of Shore Pods ranging from 1P to 3P.

Purpose

Shore Pod developers should play an active role in ensuring the success of Shore Pod missions by implementing good engineering practice, testing, and verification of their systems.

Shore Pod Specification

This section presents basic general requirements that shall be applied on all Shore Pods as well as all mechanical dimensions.

Mechanical Requirements

Shore Pods are sensor platform buoys based on a cylindric-shaped form factor (approximately 8cm (OD) x 12 (cm) (h) per unit). The external dimensions and features are outlined in FIG. Shore Pod’s external dimensions are slightly different depending on the number of units for the 1S, 2S and 3S cases. If additional buoyancy is needed, there is an option to add a flotation collar to the Shore Pod. The Shore Pod and Flotation Collar dimensions are outlined in Table 2.

	Outer Diameter (cm)	Length (cm)	Mass (kg)
1S	8	12	0.5
2S	8	24	1.0
3S	8	36	1.5

Table 2: Table of Measurements

1. The Shore Pod shall use the coordinate system as defined in Figure(x). The origin of the Shore Pod is located at the bottom center of the Shore Pod.
 1. Note: The Shore Pod configuration and physical dimensions shall conform to the appropriate section of Figure 1.
 2. Note: Extra volume for buoyancy may be available. This extra volume is shown in Figure(x), sometimes referred to as the Flotation Collar. The availability and volume dimensions are dispenser dependent.
2. The -Z Face of the Shore Pod shall be inserted first into the dispenser.
3. No components on the Shore Pod shall protrude further than xmm.
4. The Shore Pod shall have a 4cm envelope for each end cap.
5. Deployables shall be constrained by the Shore Pod, not the dispenser.

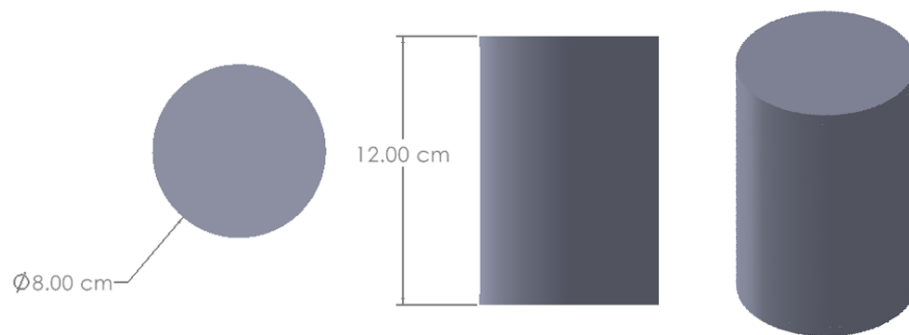


Figure 1: 1S Shore Pod Drawing.

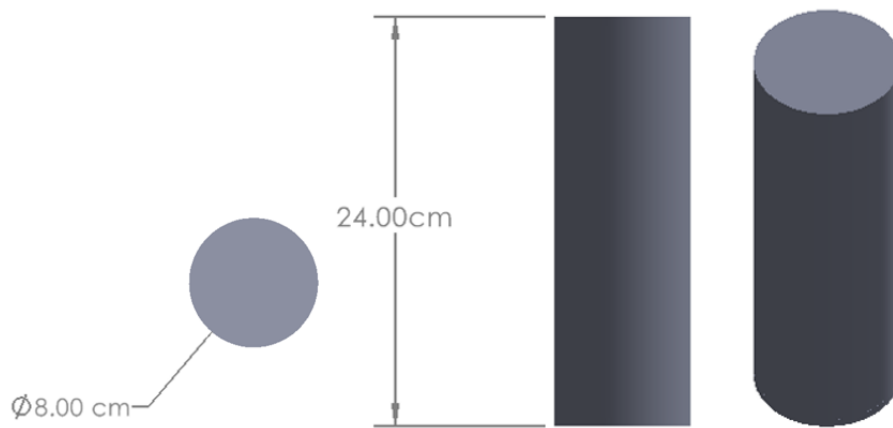


Figure 2: 2S Shore Pod Drawing.

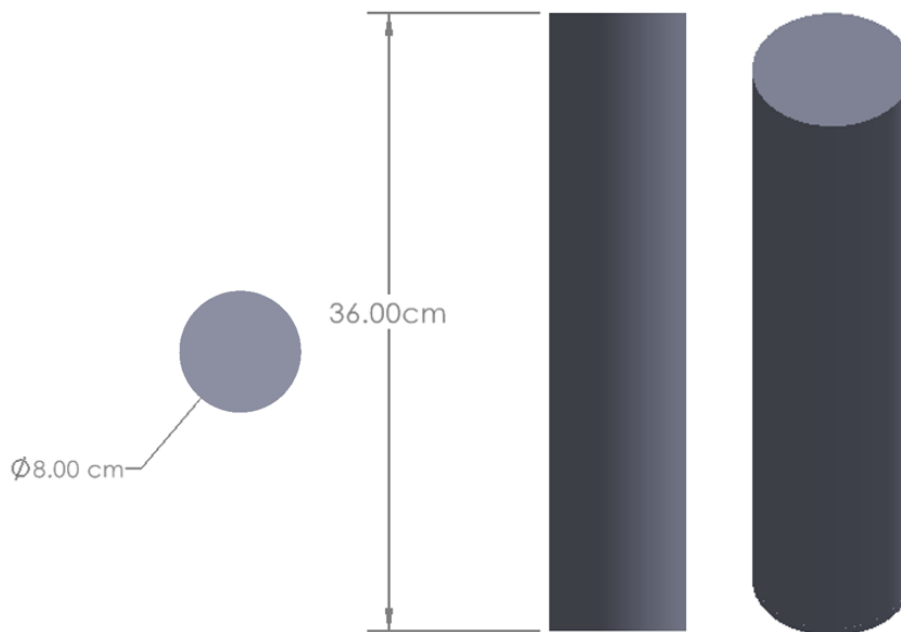


Figure 3: 3S Shore Pod Drawing.

6. Rails shall have a minimum width of xmm measured from the edge of the rail to the first protrusion on each face.
7. Rails shall have a coefficient of friction less than x.
8. The edges of the rails shall be rounded to a radius of xmm
9. At least 75% of the rail should be in contact with the dispenser rails. 25% of the rails may be recessed.
10. The Shore Pod shall have a maximum mass as defined in Table 1.
 1. Note: Masses larger than the ones presented in Table may be evaluated on a case-by-case basis. Verify your constraints with your dispenser provider.
 2. Note: Acceptable masses may vary depending on your dispenser capabilities. Verify capabilities with your dispenser provider.
11. The Shore Pod center of gravity shall fall within the ranges specified in Table.
12. The Shore Pod should be made from material that falls into the density ranges provided in Table.
 1. Note: Because of the environment, the material is subject to corrosion, biofouling, water absorption and marine growth.
 - 2.

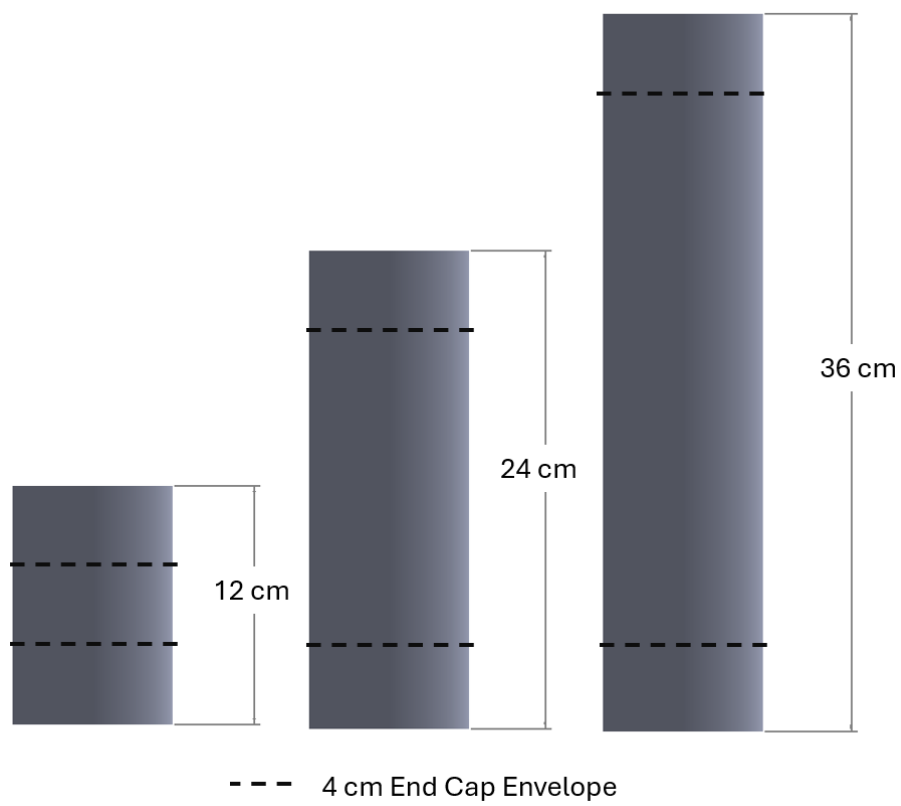


Figure 4: Shore Pod Configurations with End Cap Envelope.

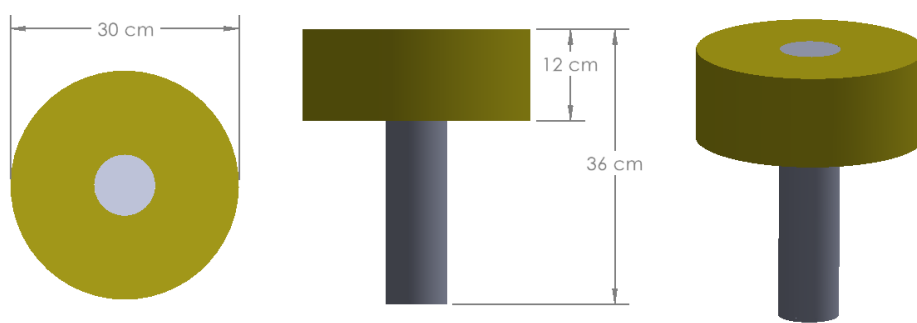


Figure 5: Example of a 30 cm collar on a 3S Shore Pod.

Electrical Specifications

1. The Shore Pod shall have at a minimum, one deployment switch, which is actuated while integrated in the dispenser.
2. The Shore Pod shall protect against electrical issues due to water.
- 3.

Operational Specifications

1. Radio Frequencies in ISM band shall comply with ISM requirements.
2. Operators shall obtain and provide documentation of proper licenses for use of radio frequencies.
 1. Note: For amateur frequency use, this requires proof of frequency coordination by the IARU. Applications can be found at www.iaru.org.
3. The Shore Pod shall comply with their country's radio license agreements and restrictions.
 1. Note: Shore Pod operator should refer to the International Telecommunication Union (ITU) to determine what licenses and approvals are needed for their country.
4. All deployables such as booms, antennas, and solar panels shall wait to deploy a minimum of 1 minute after the Shore Pod deployment switch(es) are activated during deployment.